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Kim

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(54) **METHOD OF INTERWORKING BETWEEN SPECTRUM SHARING SYSTEM AND DISTRIBUTED ANTENNA SYSTEM**

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(30) **Foreign Application Priority Data**

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(57) **ABSTRACT**

The disclosure provides a method of operating a distributed antenna system (DAS) interworking with a spectrum sharing system (SSS) including: transmitting, by a node unit of the DAS, DAS information to a management system entity (MSE); generating, by the MSE, linkage information based on the DAS information and radio service device (RSD) information received from at least one RSD of the SSS; transmitting, by the MSE, the interworking information to a system controller of the SSS; receiving, by the MSE, allocation information including a result of allocating shared radio resources to the DAS and the at least one RSD, respectively, according to the interworking information from the system controller; transmitting, by the MSE, the allocation information to the node unit; and operating, by the node unit, according to the allocation information.

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(52) **U.S. Cl.**

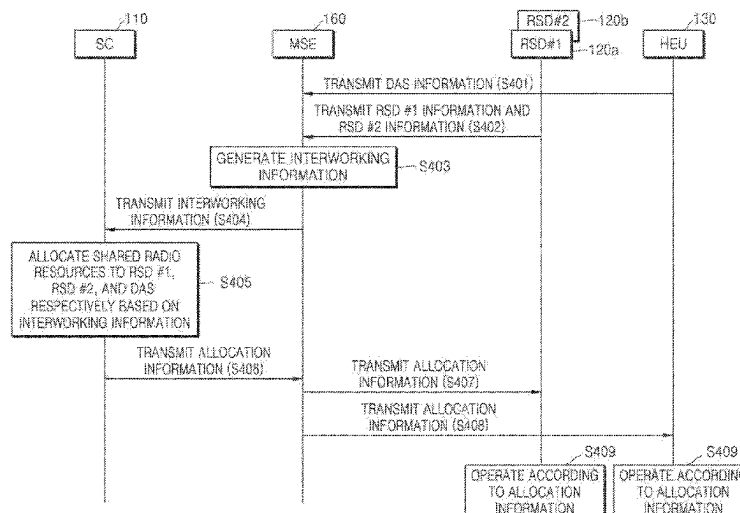
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(58) **Field of Classification Search**

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17 Claims, 16 Drawing Sheets



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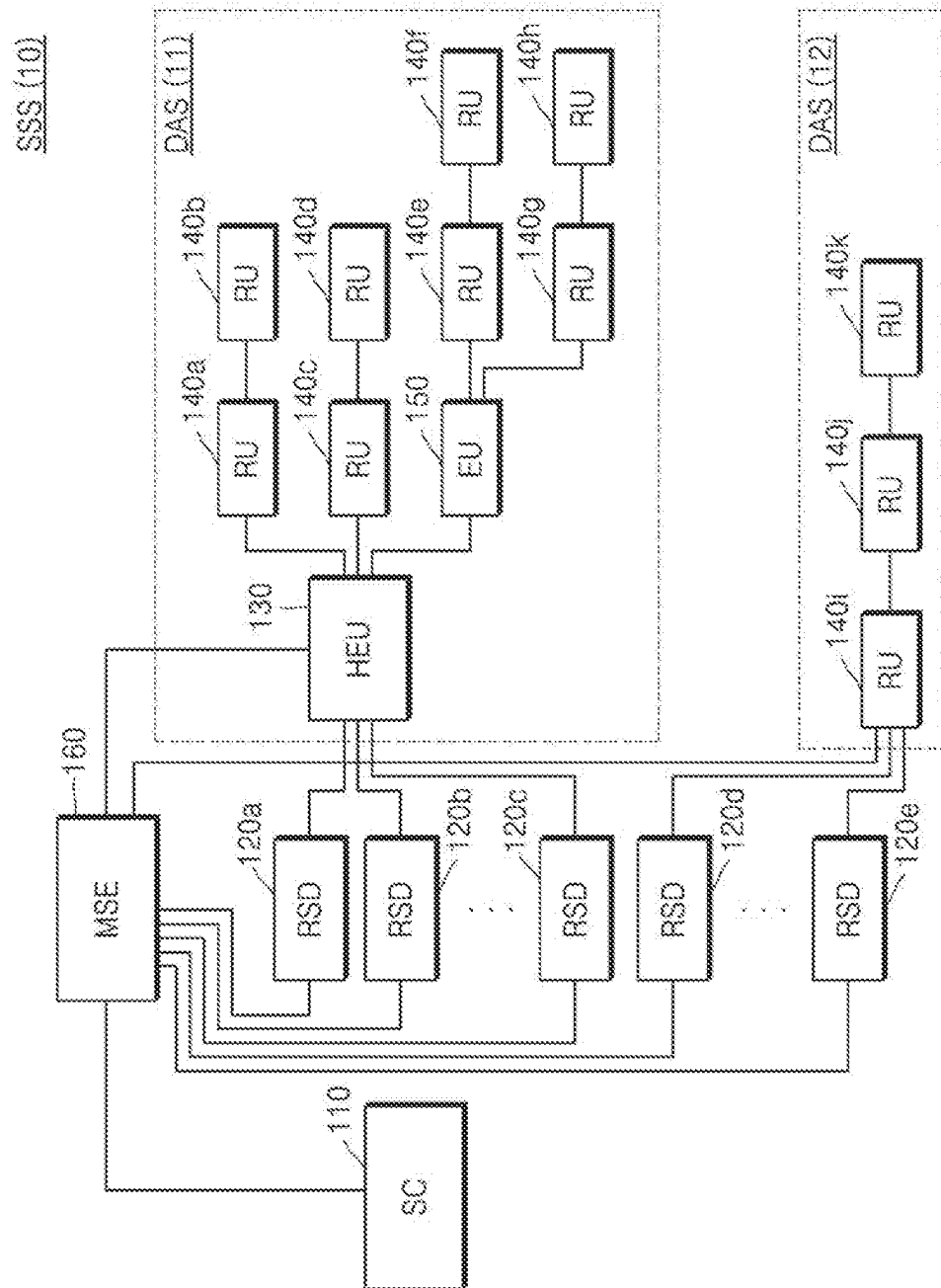


FIG. 2A

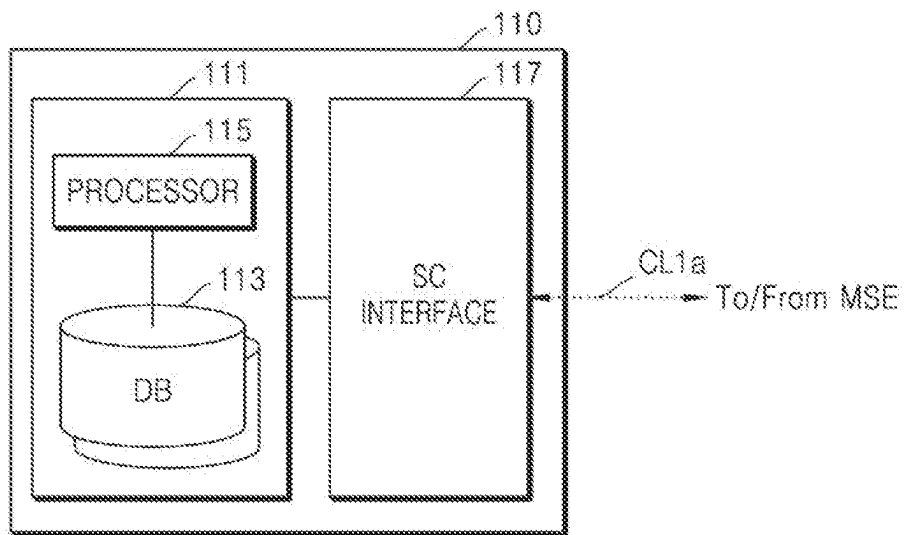


FIG. 2B

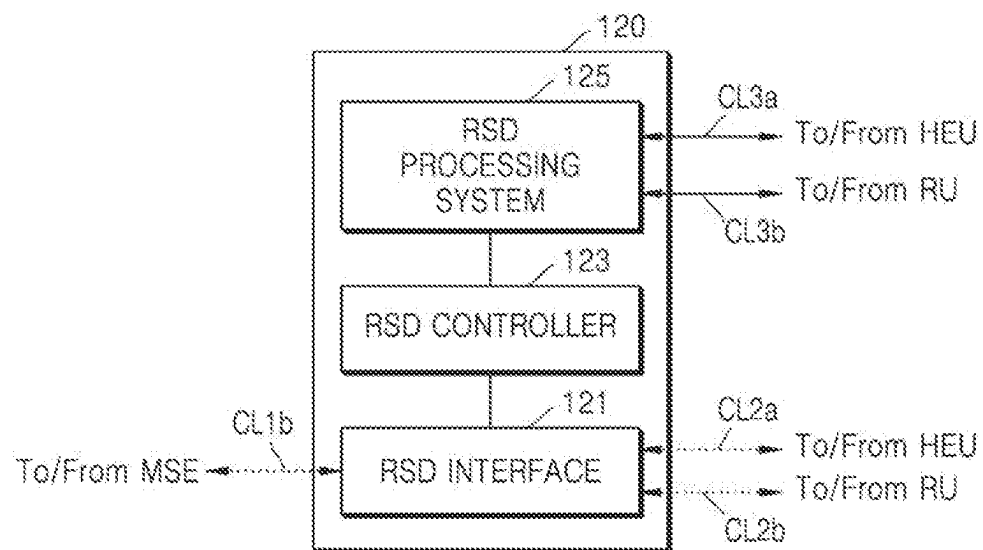


FIG. 2C

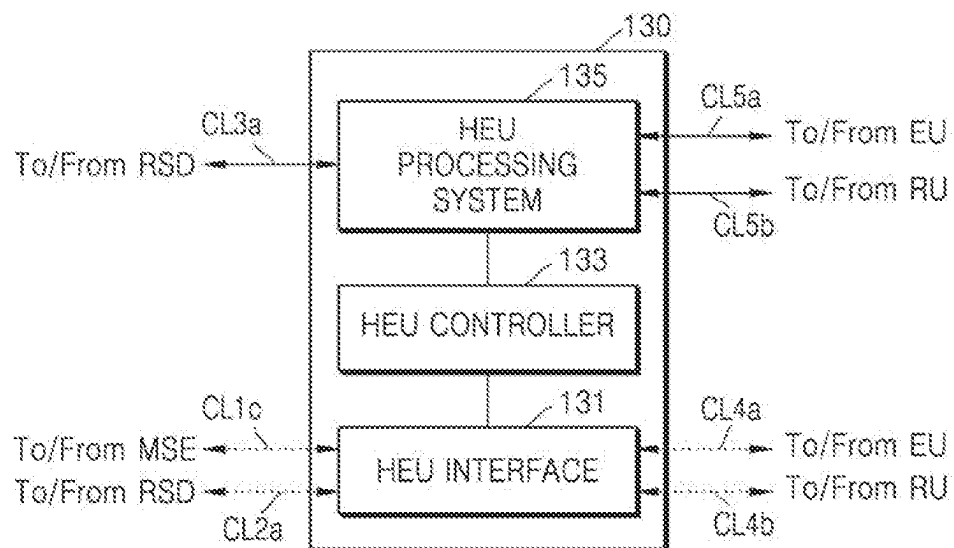


FIG. 2D

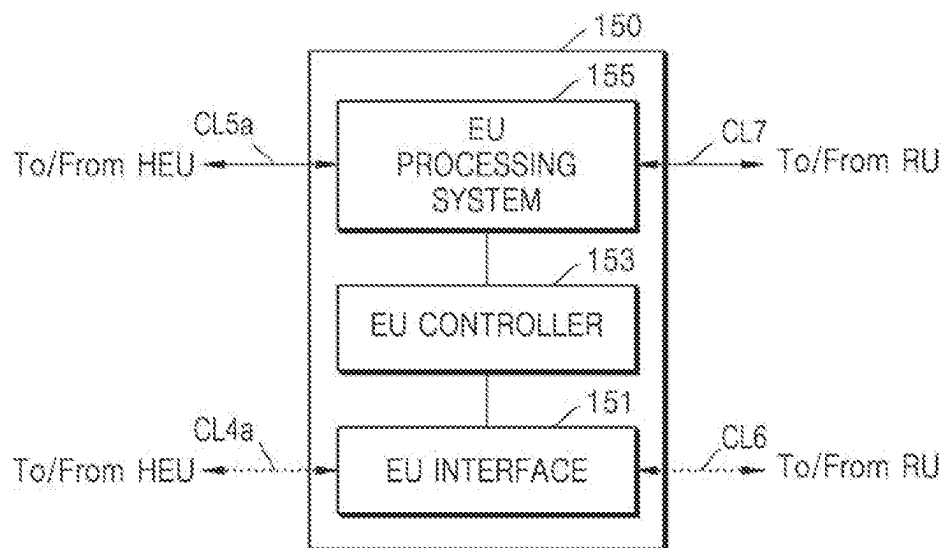


FIG. 2E

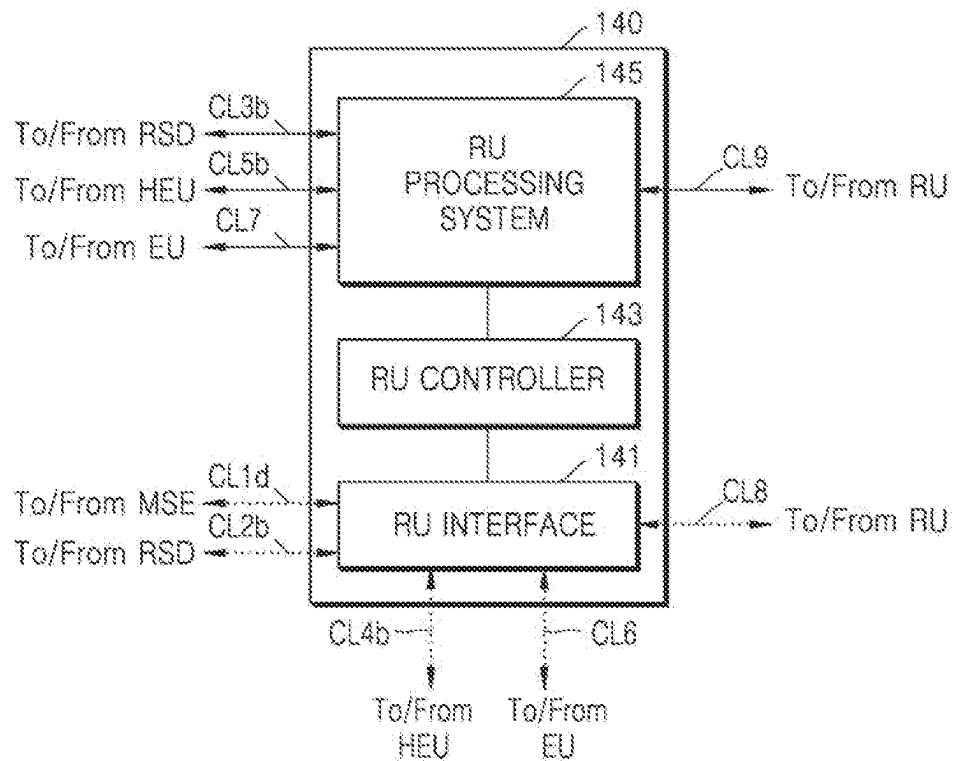


FIG. 2F

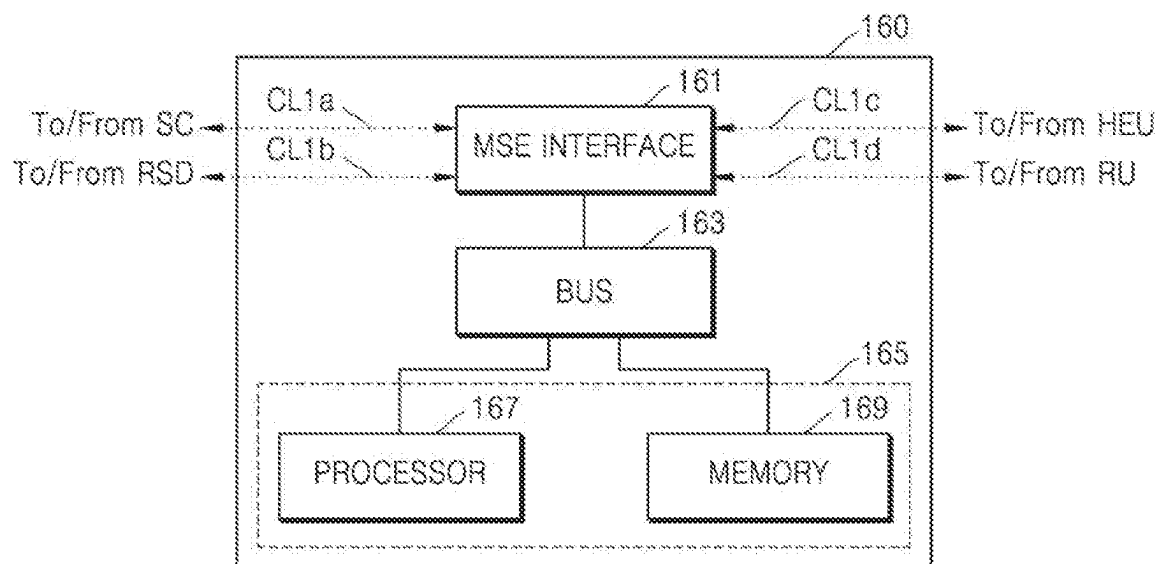


FIG. 3

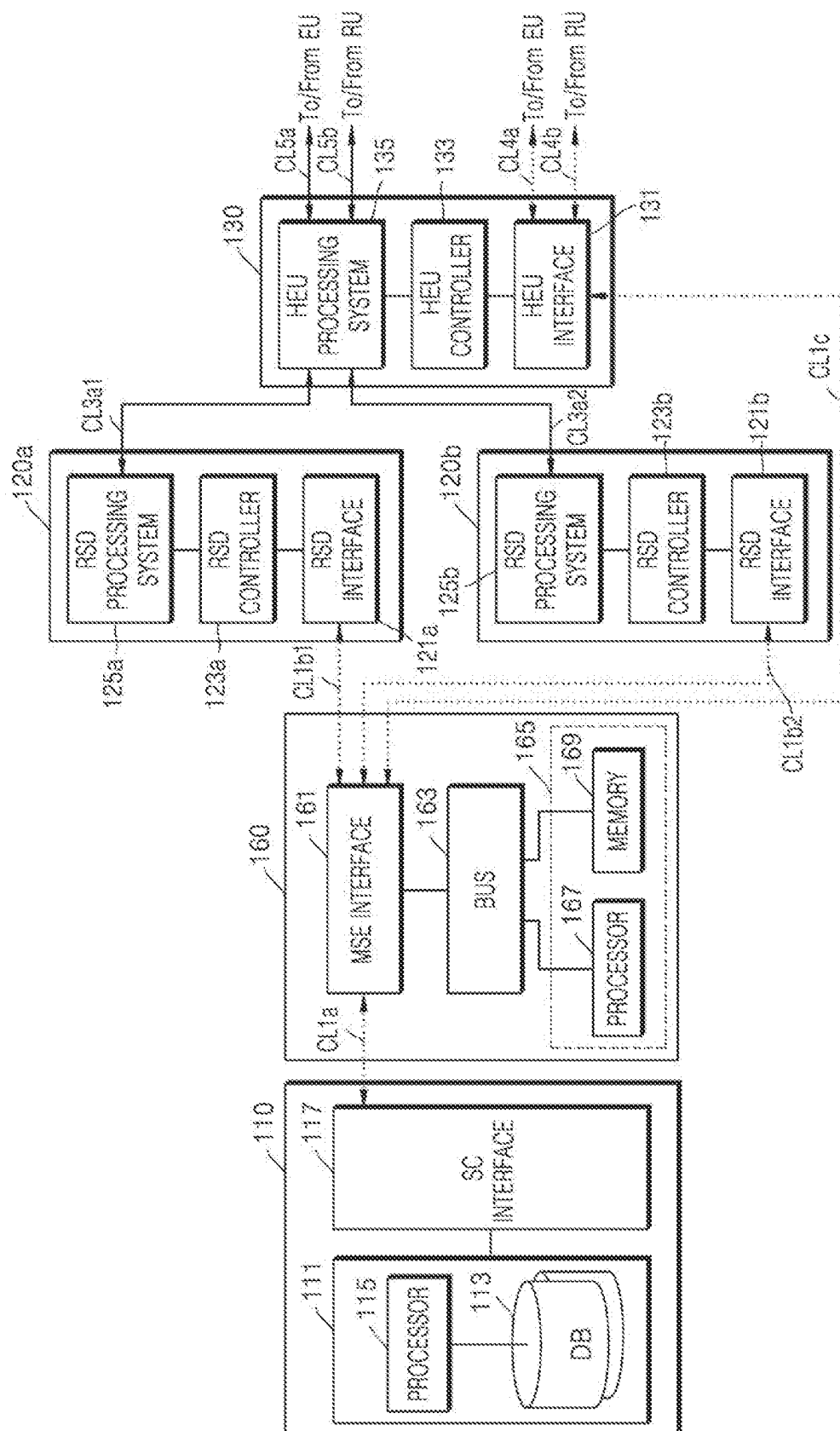


FIG. 4

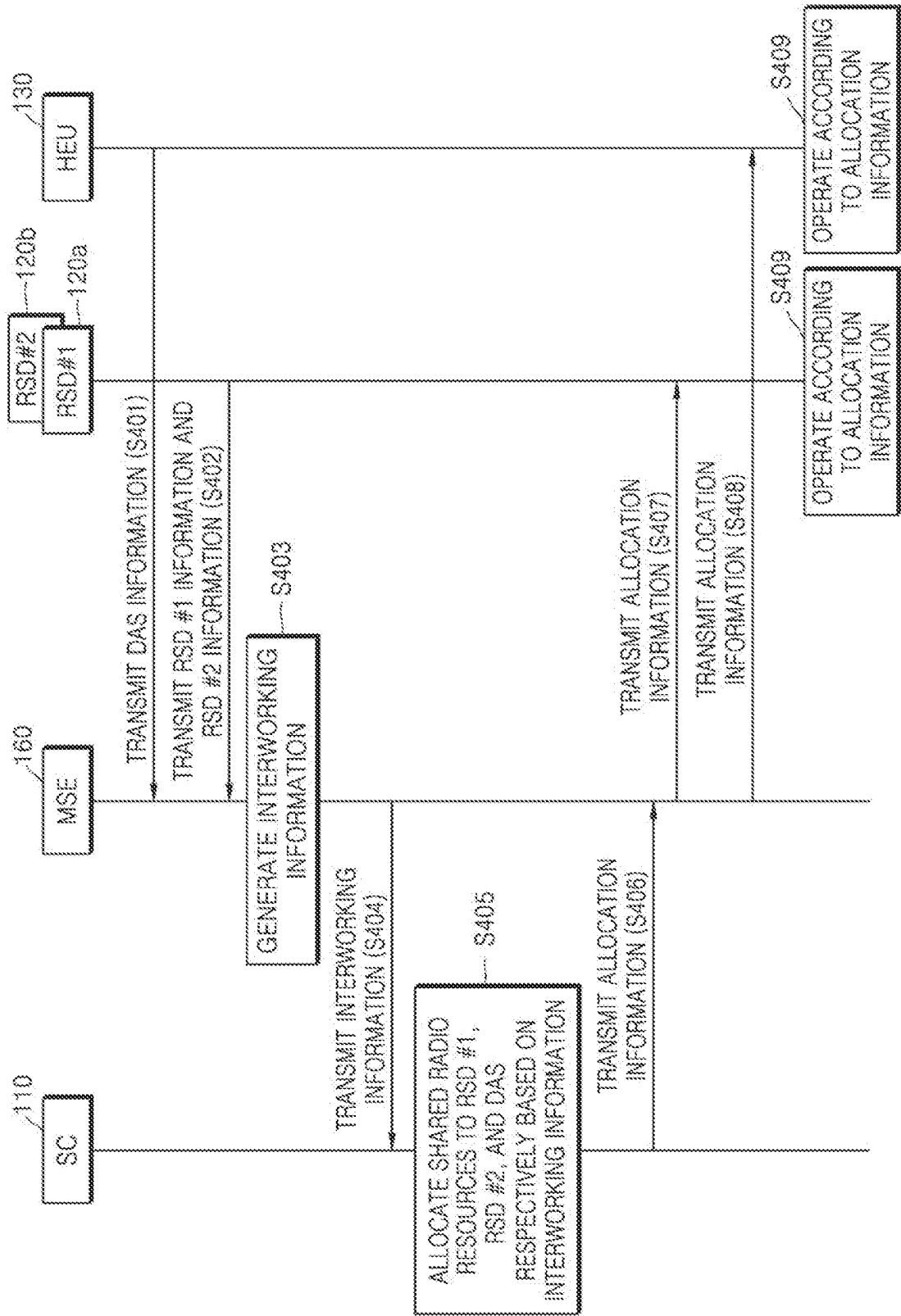


FIG. 5

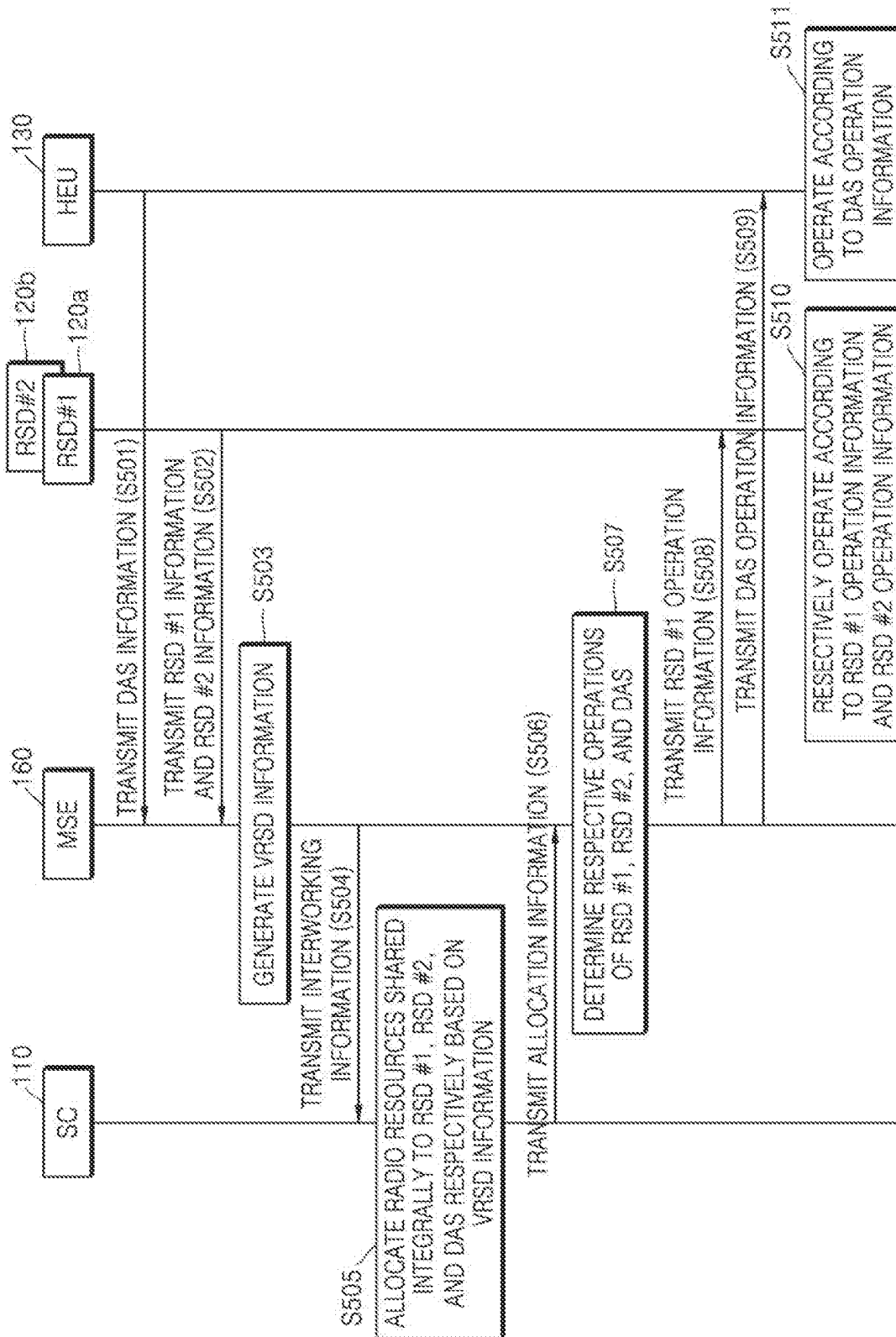


FIG. 6

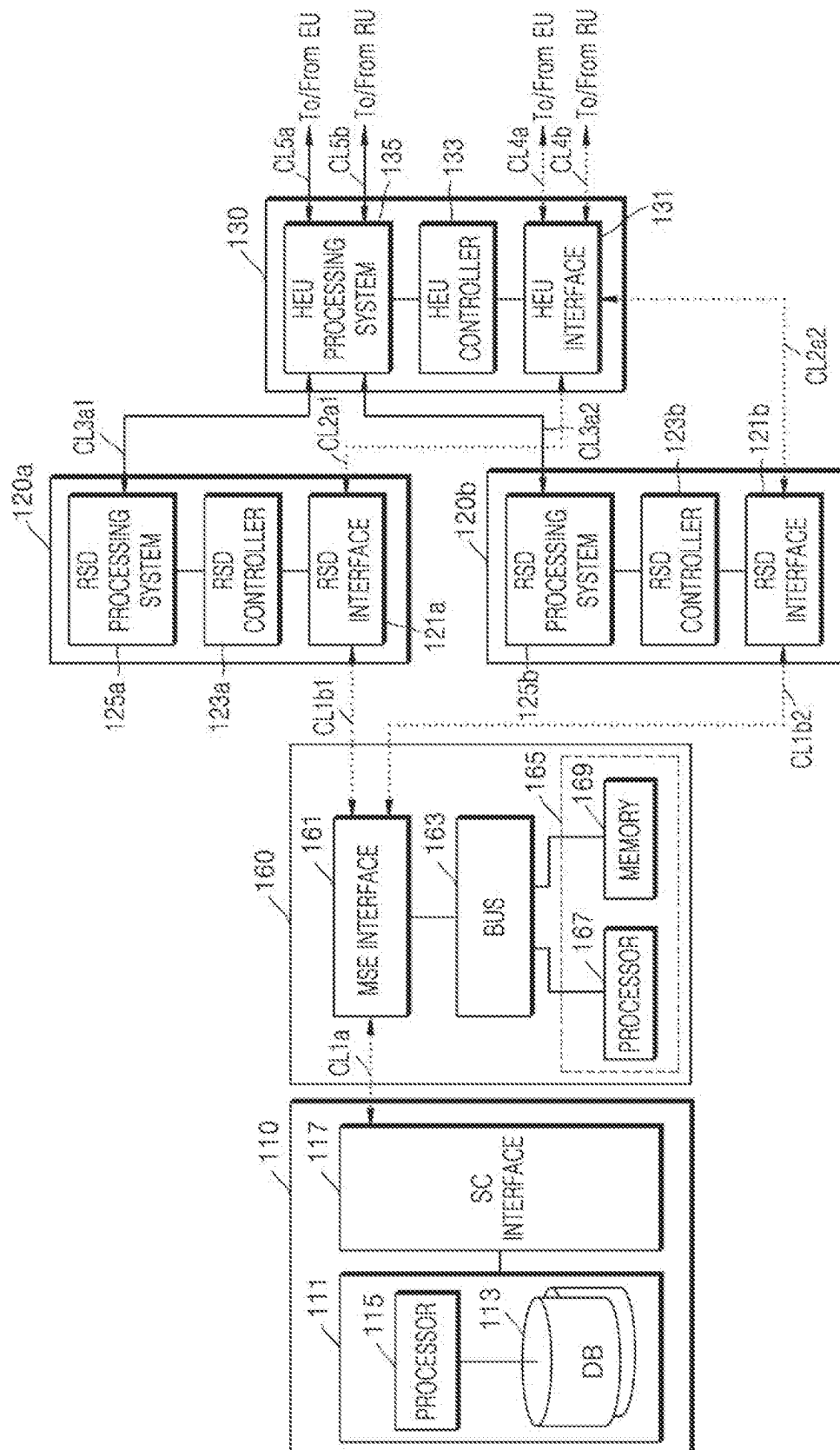


FIG. 8

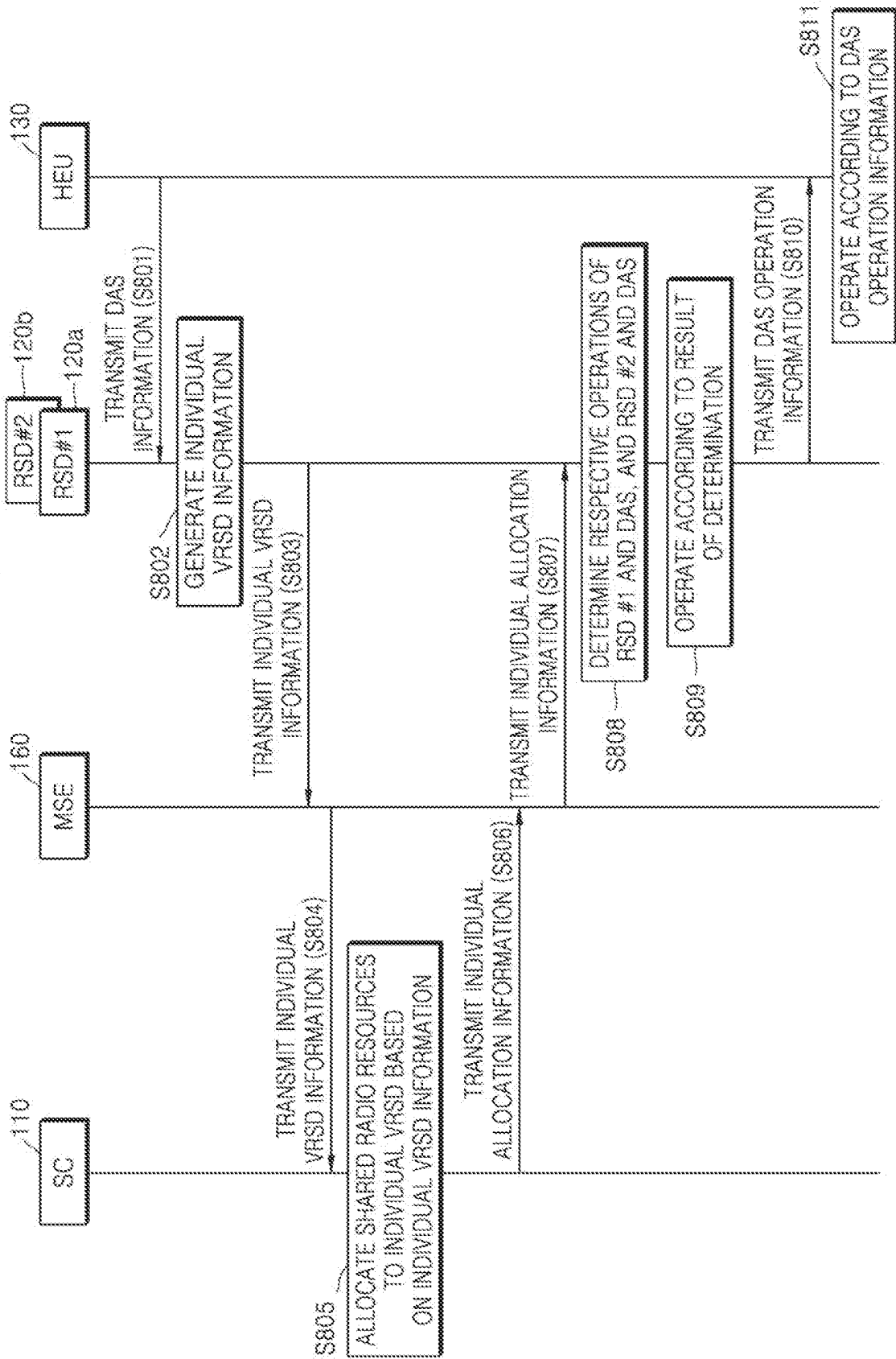


FIG. 9

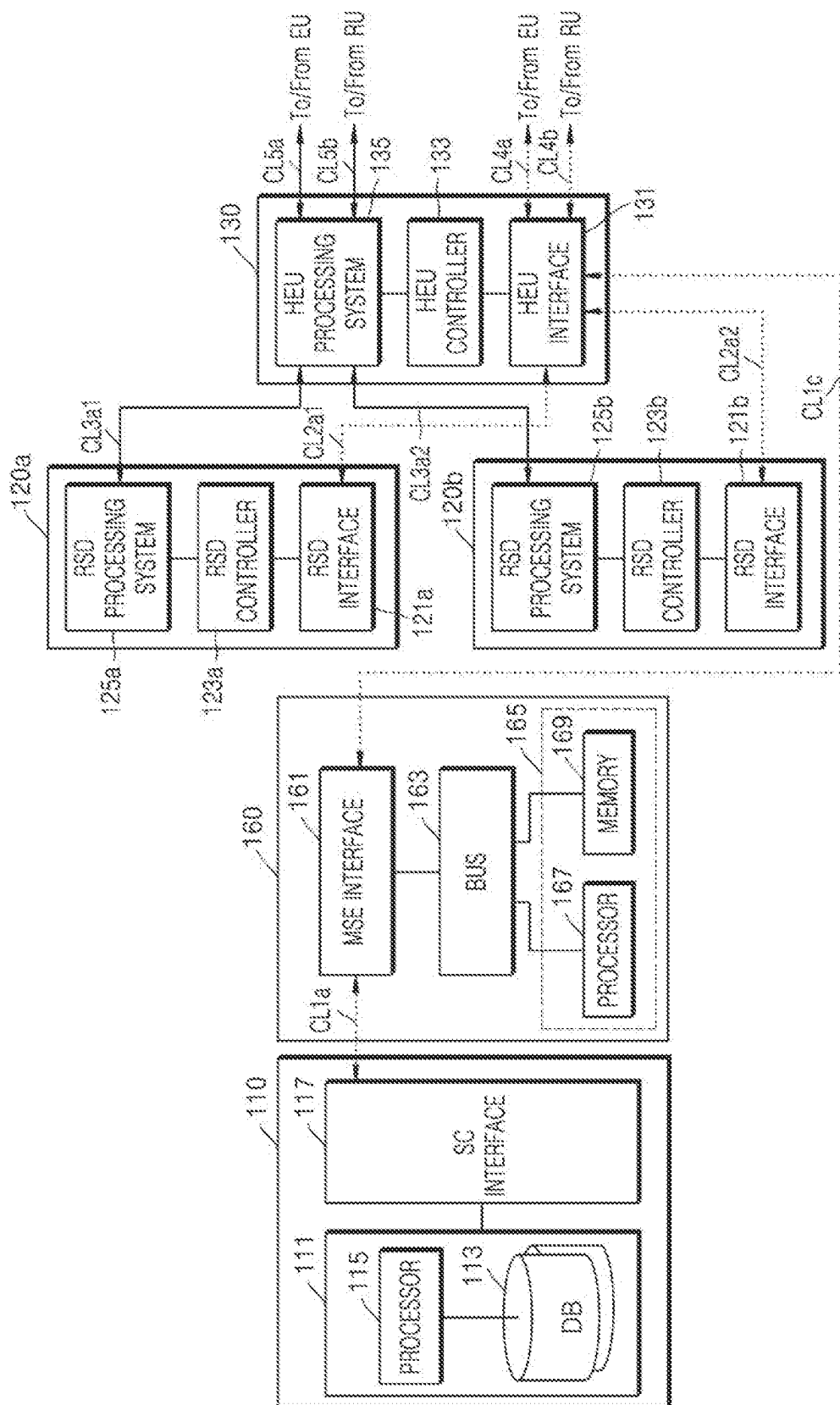
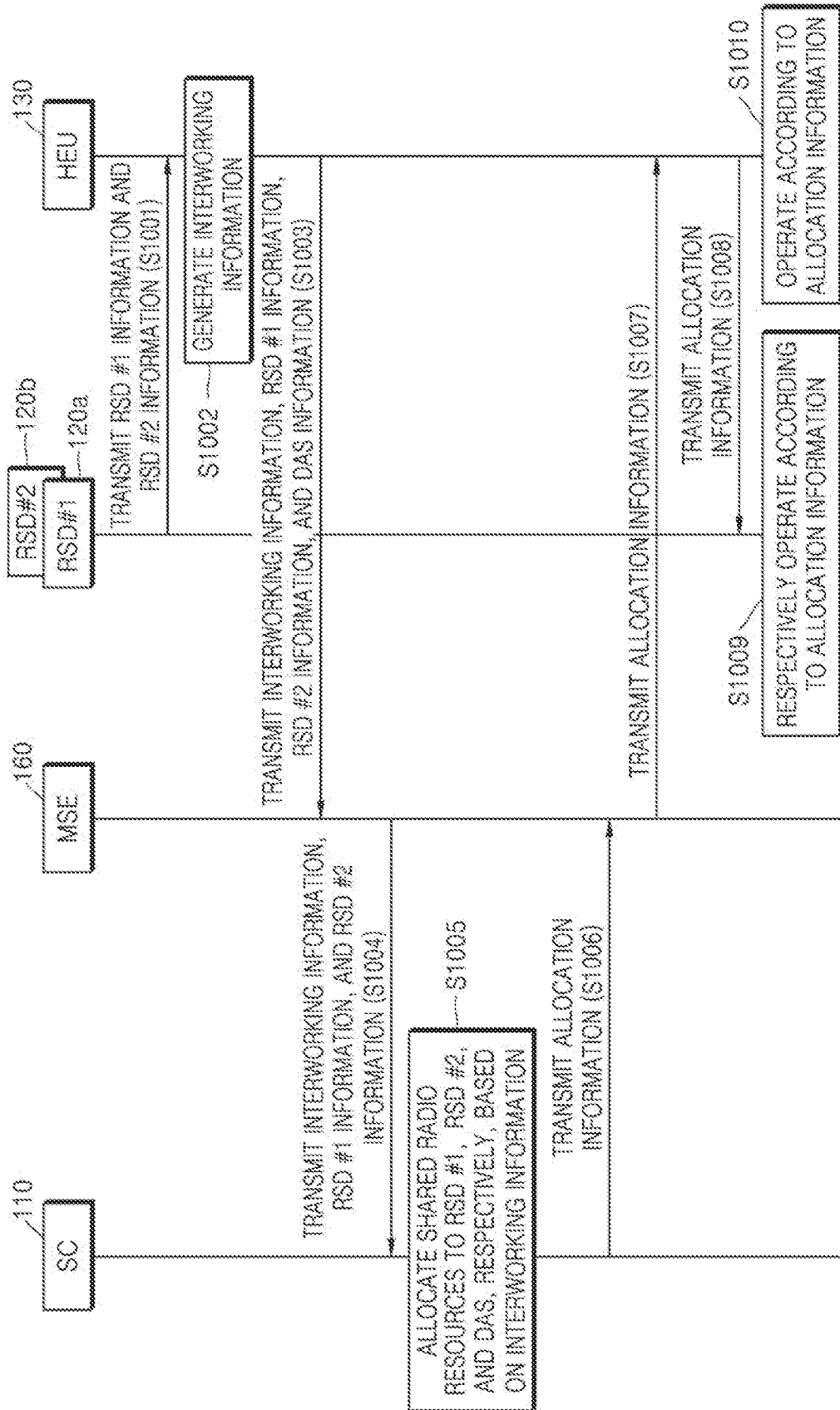


FIG. 10



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METHOD OF INTERWORKING BETWEEN SPECTRUM SHARING SYSTEM AND DISTRIBUTED ANTENNA SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. application Ser. No. 17/348,136 filed on Jun. 15, 2021, which claims the benefit of Korean Patent Application No. 10-2020-0073272, filed on Jun. 16, 2020, in the Korean Intellectual Property Office, the disclosures of which are incorporated herein in its entirety by reference.

BACKGROUND

1. Field

One or more embodiments relate to a method of interworking between a spectrum sharing system and a distributed antenna system.

2. Description of the Related Art

In order to cope with the increasing demand of mobile traffic and the limitation of frequency spectrum (or spectrum) retrieval and relocation, the introduction of radio station management and a service system based on spectrum sharing is being actively discussed in order to efficiently utilize limited radio resources (e.g., a bandwidth and transmission power) mainly in major advanced countries.

For example, the United States has announced the introduction of Citizens Broadband Radio Service (CBRS), which is an urban spectrum sharing service in the 3.5 GHz band. In addition, the United Kingdom has announced the introduction of spectrum co-use for the 3.8 GHz to 4.2 GHz bands based on the Framework for Spectrum Sharing.

Such a spectrum sharing service is expected not only to be applied in the existing specific service field but also to provide a sufficient advantage for substituting and converging various services as well as supplementing a mobile communication service such as 5G.

A distributed antenna system (DAS) is a transmission medium such as optical fiber, wired Ethernet, and the like, or a system composed of spatially separated antenna nodes (e.g., remote units) connected to a common node (e.g., a head-end unit) through a transmission network.

The DAS is installed in an area where radio signals are not received or where radio signals are weak, such as inside buildings, underground buildings, subways, tunnels, apartment complexes in a residential area, stadiums, and the like to extend coverage of a base station by providing communication services to even a shadow area where signals of the base station are difficult to reach.

The DAS is closely related to a neutral host radio access network model proposed by the CBRS Alliance, and is likely to interwork with the spectrum sharing system or to be applied as a part of the spectrum sharing system.

However, a concrete method of interworking between the DAS and the spectrum sharing system has not been proposed yet.

SUMMARY

One or more embodiments include a method of interworking between a spectrum sharing system and a distributed antenna system.

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The disclosure is not limited to the above objectives, but other objectives not described herein may be clearly understood by those of ordinary skilled in the art from descriptions below.

According to an aspect of the disclosure, there is provided a method of operating a distributed antenna system (DAS) interworking with a spectrum sharing system (SSS), the method includes: transmitting, by a node unit of the DAS, DAS information to a management system entity (MSE); generating, by the MSE, interworking information based on the DAS information and radio service device (RSD) information received from at least one RSD of the SSS; transmitting, by the MSE, the interworking information to a system controller of the SSS; receiving, by the MSE, allocation information including a result of allocating shared radio resources to the DAS and the at least one RSD, respectively, according to the interworking information from the system controller; transmitting, by the MSE, the allocation information to the node unit; and operating, by the node unit, according to the allocation information.

According to an exemplary embodiment, the interworking information may include information about at least two of an indication of an interworking state of the at least one RSD and the DAS, an indication of radio access technology (RAT) provided by the at least one RSD through the DAS, operation parameters related to the RAT, a geographic location, an available channel, and an available frequency spectrum range.

According to an exemplary embodiment, the method may further include, transmitting, by the MSE, the allocation information to the at least one RSD after the receiving of the allocation information.

According to an exemplary embodiment, the transmitting of the interworking information may include, transmitting, by the MSE, the interworking information to the system controller as part of a registration process for the system controller of the DAS and the at least one RSD.

According to an exemplary embodiment, the transmitting of the interworking information may include, transmitting, by the MSE, the interworking information to the system controller through at least one of a resource request to the system controller or periodic status update of the DAS and the at least one RSD.

According to an exemplary embodiment, the MSE may be configured to operate and manage the DAS by controlling the operation of the node unit.

According to an exemplary embodiment, the MSE may be configured to control an operation of the at least one RSD to operate and manage the SSS.

According to an exemplary embodiment, the node unit may be a head-end unit of the DAS communicatively connected to the at least one RSD.

According to an exemplary embodiment, the node unit may be a remote unit of the DAS communicatively connected to the at least one RSD.

According to an aspect of the disclosure, there is provided a method operating a distributed antenna system (DAS) interworking with a spectrum sharing system (SSS), the method includes: transmitting, by a node unit of the DAS, DAS information to a management system entity (MSE); generating, by the MSE, virtualized RSD information based on the DAS information and radio service device (RSD) information received from at least one RSD of the SSS; transmitting, by the MSE, the virtualized RSD information to a system controller of the SSS; receiving, by the MSE, allocation information including a result of allocating radio resources integrally shared to the DAS and the at least one

RSD according to the virtualized RSD information from the system controller; determining, by the MSE, an operation of the DAS based on the allocation information; and operating, by the node unit, according to the determined operation.

According to an exemplary embodiment, the virtualized RSD information may be information that causes the system controller of the SSS to recognize the at least one RSD and the DAS as one device or the DAS as an extension device of the at least one RSD.

According to an exemplary embodiment, the virtualized RSD information may include information about at least two of an indication of a radio access technology (RAT) integrally supported by the at least one RSD and the DAS, operation parameters related to the RAT, a geographic location, an available channel, and an available frequency spectrum range.

According to an exemplary embodiment, the determining of the operation of the DAS may include, determining, by the MSE, an operation of the at least one RSD based on the allocation information.

According to an exemplary embodiment, the transmitting of the virtualized RSD information may include, transmitting, by the MSE, the virtualized RSD information to the system controller as part of a registration process for the system controller of the DAS and the at least one RSD.

According to an exemplary embodiment, the transmitting of the virtualized RSD information may include, transmitting, by the MSE, the virtualized RSD information to the system controller through at least one of a resource request to the system controller or periodic status update of the DAS and the at least one RSD.

According to an aspect of the disclosure, there is provided a method operating a distributed antenna system (DAS) interworking with a spectrum sharing system (SSS), the method includes: transmitting, by a node unit of the DAS, DAS information to the at least one RSD so that at least one radio service device (RSD) of the SSS generates interworking information or virtualized RSD information; receiving, by a management system entity (MSE), the interworking information or the virtualized RSD information from the at least one RSD; transmitting, by the MSE, the interworking information or the virtualized RSD information to a system controller of the SSS; receiving, by the MSE, allocation information including a result of allocating shared radio resources according to the interworking information or the virtualized RSD information from the system controller; transmitting, by the MSE, the allocation information to the at least one RSD; and operating, by the node unit, under the control of the at least one RSD.

According to an aspect of the disclosure, there is provided a method operating a distributed antenna system (DAS) interworking with a spectrum sharing system (SSS), the method includes: receiving, by a node unit of the DAS, radio service device (RSD) information from at least one RSD of the SSS; generating, by the node unit, interworking information or virtualized RSD information based on DAS information and the RSD information; transmitting, by the node unit, the interworking information or the virtualized RSD information to a management system entity (MSE); transmitting, by the MSE, the interworking information or the virtualized RSD information to a system controller of the SSS; receiving, by the MSE, allocation information including a result of allocating shared radio resources according to the interworking information or the virtualized RSD information from the system controller; transmitting, by the

MSE, the allocation information to the node unit; and operating, by the node unit, according to the allocation information.

BRIEF DESCRIPTION OF THE DRAWINGS

These and/or other aspects will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings in which:

FIG. 1 is a block diagram of a spectrum sharing system interworking with a distributed antenna system according to an embodiment;

FIGS. 2A to 2F are block diagrams of elements of a spectrum sharing system according to an embodiment;

FIG. 3 is a block diagram of a spectrum sharing system according to an embodiment, and FIGS. 4 and 5 are flowcharts for illustrating a method of operating the spectrum sharing system shown in FIG. 3;

FIG. 6 is a block diagram of a spectrum sharing system according to an embodiment, and FIGS. 7 and 8 are flowcharts for illustrating a method of operating the spectrum sharing system shown in FIG. 6; and

FIG. 9 is a block diagram of a spectrum sharing system according to an embodiment, and FIGS. 10 and 11 are flowcharts for illustrating a method of operating the spectrum sharing system shown in FIG. 9.

DETAILED DESCRIPTION

An example of a spectrum sharing system of the disclosure is a Citizens Broadband Radio Service (CBRS) system specified by the United States Federal Communications Commission (FCC). Hereinafter, for convenience of description, technologies proposed in the disclosure will be described on the premise of the CBRS system. However, such a description does not limit that the technologies proposed in the disclosure are applied to various spectrum sharing systems (e.g., Licensed Spectrum Access (LSA) system specified by Europe) other than the CBRS system.

The spectrum sharing system of the disclosure is a new type of system in which two or more wireless communication systems provide authorized shared access in conjunction with an in-building wireless communication system (e.g., a distributed antenna system (DAS)), which is further developed from a general CBRS system that provides or participates in authorized shared access between two or more wireless communication networks or two or more wireless communication systems (e.g., citizens broadband service devices (CBSDs) or CBSD domain proxies).

As the spectrum sharing system of the disclosure operates with the in-building wireless communication system, such as a distributed antenna system, as an element, it is required to protect radio resources from each other based on constraints due to radio access technologies being used by the in-building wireless communication system, as well as radio access technologies (RATs) being used by general competing users or wireless communication systems, and a plurality of operating modes for the RATs.

In a case of the DAS implemented with neutral host architecture, various radio services are integrated and provided within a service coverage. This is because various problems such as interference may be caused when the radio resources are shared without considering interworking (or interoperating) of the DAS in the spectrum sharing system.

In order to meet these requirements and to allow for optimization of radio resource allocations, various aspects of

the disclosure suggest technologies that allow system controllers of the spectrum sharing system to directly or indirectly recognize whether CBSDs (or CBSD domain proxies) interworks (or interoperates) with a DAS, and to optimize the allocation of radio resources to the CBSDs and the DAS based on a result of the recognition of interworking.

In particular, various aspects of the disclosure suggest technologies that allow the system controllers of the spectrum sharing system to receive certain information indicating an interworking status of the CBSDs (or CBSD domain proxies) and the DAS from a management system entity, to use the received information to recognize whether the CBSDs (or CBSD domain proxies) and the DAS are interworking with each other, and to optimize the allocation of radio resources to the CBSDs and the DAS based on a result of the recognition of interworking.

In various embodiments, the technologies described in the disclosure and systems and devices for implementation thereof may utilize RATs such as WiFi or WiMax as well as RATs such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal FDMA (OFDMA), single-carrier FDMA (SC-FDMA), LTE, a global system for mobile communications (GSM), 5G NR, and the like to support shared access to the radio spectrum between networks (or systems).

Various other embodiments and features according to the disclosure will be further described later below. It should be apparent that the teachings herein may be implemented in a wide variety of forms and any particular structure, function, or both, disclosed herein are merely exemplary, and not limiting. Based on the teachings herein, one of ordinary skill in the art will appreciate that aspects disclosed herein may be implemented independently of any other aspects, and two or more of these aspects may be combined in various ways. For example, a device or a method may be implemented by using any number of aspects set forth herein. Furthermore, the device or the method may be implemented with structures and functions of one or more of the aspects described herein, or may be implemented by using structures and functions of other aspects. For example, the method may be implemented as part of instructions stored on a non-transitory computer-readable recording medium for execution on a system, a device, an apparatus and/or a processor, or a computer. Furthermore, one aspect may include at least one component of the claim.

Hereinafter, various embodiments of the disclosure will be described in detail in order.

FIG. 1 is a block diagram of a spectrum sharing system (SSS) 10 according to an embodiment.

The SSS 10 may include a system controller (SC) 110, radio service devices (RSDs) 120a to 120e, first and second distributed antenna systems (DAS) 11 and 12, and a management system entity (MSE) 160.

The SSS 10 provides some degree of protection to existing users (e.g., fixed satellite systems, WISPs, and government/military systems) with potentially higher priorities, other users, and radio service providers while allowing shared radio resources by control of the SC 110, for example, operating frequencies, power limits, a geographical area, or the like, to be dynamically allocated to multiple users and radio service providers related to the RSDs 120a to 120e and the first and second DAS 11 and 12.

The SC 110 may control overall spectrum sharing in the SSS 10, through the MSE 160, by accepting requests for use of the shared radio resources from the RSDs 120a to 120e and/or the first and second DAS 11 and 12, by solving

conflicts or over-constraints in these requests, and by approving the use of the shared radio resources for the radio access services.

For example, the SC 110 may receive interworking information of the RSDs 120a to 120e transmitted from the MSE 160 and a corresponding one of the first and second DAS 11 and 12 during a registration and resource request or a periodic status update process from among operation procedures for allocation and reallocation of the shared radio resources. The SC 110 may determine whether the RSDs 120a to 120e interwork with the corresponding one of the first and second DAS 11 and 12 based on the received interworking information, and allocate the shared radio resources in consideration of a result of the determination. This will be described in more detail later with reference to FIGS. 3 to 11.

The term “interworking” means that the RSDs 120a to 120e are used as signal sources of at least one of the first and second DAS 11 and 12.

In addition, the term “determining” encompasses a wide variety of actions. For example, the term “determining” may include computing, processing, deriving, examining, looking up (e.g., looking up in a table, database, or other data structure), identifying, and the like. The term “determining” may also include receiving (e.g., receiving information), accessing (accessing data in a memory), and the like. The term “determining” may also include resolving, selecting, choosing, establishing, and the like.

The RSDs 120a to 120e may be devices that provide radio services using any radio access technology, such as a base station, an access point, or any type of radio frequency (RF) access system.

The RSDs 120a to 120e may be communicatively connected to the MSE 160 and/or a node unit of any one of the first and second DAS 11 and 12. In addition, the RSDs 120a to 120e may be communicatively connected to the SC 110 through the MSE 160 and/or any one node unit of the first and second DAS 11 and 12. As the RSDs 120a to 120e are not directly connected to the SC 110 but indirectly connected to the SC 110 through the MSE 160 or the like, when the number of RSDs constituting the SSS 10 increases, it is possible to effectively reduce the burden of managing, controlling, and operating the RSDs of the SC 110.

The RSDs 120a to 120e may provide a radio service to end user devices in a cell using a spectrum allocated by indirect control of the SC 110 (e.g., control through the MSE 160, the first and second DAS 11 and 12, etc.).

Alternatively, the RSDs 120a to 120e may provide a radio service to the end user devices through a corresponding one of the first and second DAS using the spectrum allocated by the indirect control of the SC 110.

Although not shown in FIG. 1, some of the RSDs 120a to 120e may be communicatively connected to the SC 110 and may provide a radio service to the end user devices in the cell directly or through a corresponding one of the first and second DAS 11 and 12 by using a spectrum allocated by direct control of the SC 110.

In addition, although not shown in FIG. 1, some of the RSDs 120a to 120e may be connected to other RSDs, and thus may each function as a domain proxy for lower RSDs.

Each of the first and second DAS 11 and 12 may combine/distribute a radio service provided from at least one corresponding one of the RSDs 120a to 120e by the indirect control of SC 110 (control through the MSE 160 or the at least one corresponding RSD of the RSDs 120a to 120e), and may provide the radio service to end user devices within coverage.

According to an embodiment, the first DAS **11** may include a head-end unit (HEU) **130** connected to the RSDs **120a**, **120b**, and **120c** and/or the MSE **160**, remote units (RU) **140a** and **140c** connected to the HEU **130** in a point-to-multipoint structure, and RUs **140b** and **140d** respectively connected to corresponding ones of the RUs **140a** and **140c** in a daisy chain structure.

As shown in FIG. **1**, the first DAS **11** may further optionally include an expansion unit (EU) **150**, and RUs **140e** to **140h** may be connected to the EU **150** in a mixed form of the point-to-multipoint structure and the daisy-chain structure.

The first DAS **11** may provide radio services from the RSDs **120a**, **120b**, and **120c** to the end user devices by using a radio resource allocated according to the indirect control of the SC **110** (e.g., control through the MSE **160** or the RSDs **120a**, **120b**, and **120c**).

According to an embodiment, the second DAS **12** may include an RU **140i** connected to the RSDs **120d** and **120e** and/or the MSE **160**, and RUs **140j** and **140k** connected to the RU **140i** in a daisy chain structure.

The RUs **140i**, **140j**, and **140k** may process a plurality of radio services in an integrated manner, unlike a remote radio head, which is an RF processing device of a distributed base station. FIG. **1** shows only an embodiment in which the RUs **140i**, **140j**, and **140k** are directly connected to the RSDs **120d** and **120e**, but the RUs **140i**, **140j**, and **140k** may also be connected to the RSDs **120d** and **120e** through a certain network.

The second DAS **12** may provide radio services from the RSDs **120d** and **120e** to the end user devices by using the radio resource allocated according to the indirect control of the SC **110** (e.g., control through the MSE **160** or the RSDs **120d**, **120e**).

The MSE **160** may be communicatively connected to the SC **110** and the RSDs **120a** to **120e** and/or the node unit of each of the first and second DAS **11** and **12**.

According to an embodiment, the MSE **160** may generate interworking information indicating an interworking state based on certain information transmitted from the RSDs and the node units of the first and second DAS **11** and **12**, and the like, and may transmit the interworking information and the like to the SC **110** so that shared radio resources are allocated according to the interworking information.

According to an embodiment, the MSE **160** may receive interworking information and the like from the RSDs **120a** to **120e** and the node units of the first and second DAS **11** and **12**, and may transmit the received interworking information and the like to the SC **110** so that shared radio resources are allocated according to the received interworking information and the like.

The MSE **160** may receive allocation information including a result of the allocation of the shared radio resources from the SC **110**, and may transmit the received allocation information to the RSDs **120a** to **120e** and/or to the node units of the first and second DAS **11** and **12**.

The MSE **160** may be a system for integrated management and operation of the RSDs **120a** to **120e** and the first and second DAS **11** and **12** as described above, but is not limited thereto. In an embodiment, the MSE **160** may be a network management system or a DAS management system provided by a manufacturer of the first and second DAS **11** and **12**. In another embodiment, the MSE **160** may be a network management system provided by a manufacturer of the SC **110** and the RSDs **120a** to **120e**.

On the other hand, according to the disclosure, elements of the SSS **10**, that is, the node units (HEU, RU, and EU)

constituting the SC, RSD, and DAS, the number of MSEs, and a topology for connecting them are not limited to the embodiment shown in FIG. **1**, and various modifications and variations are possible.

FIGS. **2A** to **2F** are block diagrams of elements of a spectrum sharing system according to an embodiment. In the description of FIGS. **2A** to **2F**, the same or corresponding reference numerals as those in FIG. **1** denote the same or corresponding elements, and therefore, repeated descriptions thereof will not be given herein.

Referring to FIGS. **1** and **2A**, the SC **110** may include a system controller processing system **111** (hereinafter referred to as an SC processing system) and a system controller interface **117** (hereinafter referred to as an SC interface).

The SC processing system **111** may control all operations of the SSS **10**. For example, the SC processing system **111** may control processing operations for a registration request of at least one RSD and/or at least one DAS communicatively connected to the SC processing system **111** through the MSE **160**, processing operations for a radio resource/authorization request, status update processing operations thereof, and the like.

In particular, the SC processing system **111**, as part of the above-described operations or as a separate operation, may check whether the RSD interworks with the DAS to reflect an interworking operation state when shared radio resources are allocated.

The SC processing system **111** may include at least one database **113** and a processor **115**.

The at least one database **113** may store rules necessary for management and operation of the SSS **10**, various information about users, for example, information about priorities (e.g., a top-level incumbent user, a priority access authorized user, a general access authorized user), geographical location and/or time information, coverage, an maximum allowable power output level, a modulation type, interference threshold information, and so on.

The processor **115** may determine whether the RSD interworks with the DAS based on identification information of the RSD and/or the DAS (in more detail, node units of the DAS such as HEU, RU, and EU), interworking information, virtualized RSD information, and the like transmitted from the MSE **160**. Specific embodiments thereof will be described in more detail later with reference to FIGS. **3** to **10**.

The processor **115** may be connected to the database **113** and recognize a spectrum usage state, a usage amount, and the like of users having priority at specific times and/or geographical locations related to the RSD and the DAS that are determined whether to interwork with each other.

The processor **115** may allocate radio resources available for the RSD and the DAS based on a result of the recognition.

The processor **115** may transmit allocation information indicating a result of the allocation of the radio resources to an RSD **120** and/or the DAS to control the use of the shared radio resources by the RSD **120** and the DAS.

The SC processing system **111** may be communicatively connected to the MSE **160** through a first communication link **CL1a**, and may transmit and receive information for spectrum sharing access control to and from the MSE **160** through the SC interface **117**.

The SC processing system **111** may transmit and receive the information to and from the MSE **160** through the SC

interface **117** by using a security protocol such as a Hyper-Text Transfer Protocol over Secure Socket Layer (HTTPS) protocol.

The first communication link **CL1a** may be, for example, but is not limited to, the Internet, and may be any wired and/or wireless communication link such as WiMax, network optical fiber, an Ethernet-based cable, and the like.

Referring to FIGS. **1** and **2B**, the RSD **120** may include a radio service device interface **121** (hereinafter referred to as an RSD interface), a radio service device controller **123** (hereinafter referred to as an RSD controller), and a radio service device processing system **125** (hereinafter referred to as an RSD processing system).

The RSD interface **121** is for the RSD **120** to transmit and receive pieces of information necessary for spectrum sharing access to and from the MSE **160**, and HEU **130**, and an RU **140**.

The RSD **120** may transmit and receive the information to and from the MSE **160** connected to the RSD **120** through a first communication link **CL1b**, and the HEU **130** and the RU **140** connected to the RSD **120** through second communication links **CL2a** and **CL2b**, respectively, by using the RSD interface **121**.

The first communication link **CL1b** and the second communication links **CL2a** and **CL2b** may be, for example, but are not limited to, the Internet, and may be any wired and/or wireless communication link such as WiMax, network optical fiber, an Ethernet-based cable, and the like.

The RSD controller **123** may generate its own information related to a radio service or the like provided by the RSD **120**, interworking information indicating whether to interwork with the DAS, virtualized RSD information, and the like, and may transmit these information to the MSE **160**, the HEU **130**, or the RU **140** through the RSD interface **121**.

The RSD controller **123** may control the RSD processing system **125** according to allocation information or the like transmitted from the MSE **160** or from the HEU **130** and the RU **140** through the RSD interface **121**.

The RSD processing system **125** may activate a radio resource (e.g., a frequency spectrum or channel) allocated by the control of the RSD controller **123** and use the activated radio resource to generate service signals of radio access technology that the RSD **120** may support.

The RSD processing system **125** may transmit the generated service signals to the HEU **130** and the RU **140** through third communication links **CL3a** and **CL3b**.

The third communication links **CL3a** and **CL3b** may be media for transmitting analog or digital type service signals, for example, an RF cable, an optical fiber, an Ethernet-based cable, and the like. Although not shown in FIG. **2B**, the RSD processing system **125** may include converters for converting the generated service signals to correspond to the third communication links **CL3a** and **CL3b**.

Referring to FIGS. **1** and **2C**, the HEU **130** may include a head-end unit interface **131** (hereinafter referred to as an HEU interface), a head-end unit controller **133** (hereinafter referred to as an HEU controller), and a head-end unit processing system **135** (hereinafter referred to as an HEU processing system).

The HEU interface **131** is for the HEU **130** to transmit and receive pieces of information necessary for spectrum sharing access to and from the RSD **120**, the RU **140**, the EU **150**, and the MSE **160**.

The HEU **130** may transmit the above-described pieces of information to the RSD **120** by using a certain security protocol, for example, a HTTPS protocol.

The HEU **130** may transmit and receive pieces of information such as allocation information to and from the MSE **160**, the RU **140**, and the EU **150** by using the above-described security protocol or another security protocol defined by a manufacturer of the DAS.

The HEU **130** may transmit and receive the pieces of information to and from the MSE **160** connected to the HEU **130** through a first communication link **CL1c**, the RSD **120** connected to the HEU **130** through the second communication link **CL2a**, and the EU **150** and the RU **140** connected to the HEU **130** through fourth communication links **CL4a** and **CL4b**, respectively, by using the HEU interface **131**.

The fourth communication links **CL4a** and **CL4b** may be, for example, but are not limited to, the Internet, and may include any wired and/or wireless communication link such as WiMax, network optical fiber, an Ethernet-based cable, and the like.

The HEU controller **133** may generate interworking information, virtualization information, and the like indicating whether to interwork with the RSD **120**, and may transmit these information to the MSE **160** or the RSD **120** through the HEU interface **131**.

The HEU controller **133** may control the HEU processing system **135** according to allocation information transmitted from the MSE **160** or from the RSD **120** through the HEU interface **131**. The transmitted allocation information may be transmitted to the RU **140** and the EU **150** through the HEU interface **131**.

The HEU processing system **135** may receive service signals of the radio access technology from the RSD **120** through the third communication link **CL3a**. FIG. **2C** shows an embodiment in which one RSD **120** is connected to the HEU **130**. However, when a plurality of RSDs **120** are connected to the HEU **130**, the third communication link **CL3a** may be plural (see FIG. **3**, etc.).

The HEU processing system **135** may perform processes such as noise cancellation, filtering, combining, and the like, in an analog way and/or digitally using radio resources allocated by the control of the HEU controller **133** to the received service signals, and may transmit the combined service signals to the RU **140** and the EU **150** through the fifth communication links **CL5a** and **CL5b**.

The fifth communication links **CL5a** and **CL5b** may be media for transmitting analog or digital type service signals, for example, an RF cable, an optical fiber, an Ethernet-based cable, and the like. Although not shown in FIG. **2C**, the HEU processing system **135** may include converters for converting the combined service signals to correspond to the fifth communication links **CL5a** and **CL5b**.

Referring to FIGS. **1** and **2D**, the EU **150** may include an expansion unit interface **151** (hereinafter referred to as an EU interface), an expansion unit controller **153** (hereinafter referred to as an EU controller), and an expansion unit processing system **155** (hereinafter referred to as an EU processing system).

The EU interface **151** is for transmitting and receiving pieces of information necessary for spectrum sharing access to and from the HEU **130** and the RU **140**.

The EU **150** may transmit and receive the necessary pieces of information to and from the HEU **130** and the RU **140** by using a security protocol such as an HTTPS protocol or other security protocols defined by a manufacturer of the DAS.

The EU **150** may transmit and receive the necessary pieces of information to and from the HEU **130** connected to the EU **150** through the fourth communication link **CL4a**

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and the RU **140** connected to the EU **150** through a sixth communication link **CL6** by using the EU interface **151**.

The sixth communication link **CL6** may be, for example, but is not limited to, the Internet, and may include any wired and/or wireless communication link such as WiMax, network optical fiber, an Ethernet-based cable, and the like.

The EU controller **153** may control the EU processing system **155** according to allocation information of radio resources transmitted from the HEU **130** through the EU interface **151**.

The EU processing system **155** may receive combined service signals from the HEU **130** through the fifth communication link **CL5a** and perform processes such as amplification and the like on the combined service signals in an analog way and/or digitally by using the radio resources allocated by the control of the EU controller **153**. Thereafter, the EU processing system **155** may transmit the processed service signals to the RU **140** through a seventh communication link **CL7**.

The seventh communication link **CL7** may be a medium for transmitting analog or digital type service signals, for example, an RF cable, an optical fiber, an Ethernet-based cable, and the like. Although not shown in FIG. 2D, the EU processing system **155** may include a converter for converting the signal received through the fifth communication link **CL5a** into a signal suitable for processing therein and converters for converting the processed signal to correspond to the seventh communication link **CL7**.

Referring to FIGS. 1 and 2E, the RU **140** may include a remote unit interface **141** (hereinafter referred to as an RU interface), a remote unit controller **143** (hereinafter referred to as an RU controller), and a remote unit processing system **145** (hereinafter referred to as an RU processing system).

The RU interface **141** is for transmitting and receiving pieces of information necessary for spectrum sharing access to and from the MSE **160**, the RSD **120**, the HEU **130**, the EU **150**, and other RUs.

The RU **140**, according to an embodiment, may transmit and receive the pieces of information to and from the MSE **160** and the RSD **120** by using a security protocol such as an HTTPS protocol and may also transmit and receive the pieces of information to and from the HEU **130** and the EU **150** by using other security protocols besides the HTTPS protocol.

The RU **140** may transmit and receive the pieces of information to and from the MSE **160** connected to the RU **140** through a first communication link **CL1d**, and the RSD **120**, the HEU **130**, the EU **150**, and other RUs that are connected to the RU **140**, respectively, through the first communication link **CL1c**, the second communication link **CL2b**, the fourth communication link **CL4b**, the sixth communication link **CL6**, and an eighth communication link **CL8**.

The eighth communication link **CL8** may be, for example, but is not limited to, the Internet, and may include any wired and/or wireless communication link such as WiMax, network optical fiber, an Ethernet-based cable, and the like.

The RU controller **143** may control the RU processing system **145** according to allocation information of radio resources transmitted from the MSE **160**, the RSD **120**, the HEU **130**, or the EU **150** through the RU interface **141**.

The RU processing system **145** may receive a service signal from the RSD **120** through the third communication link **CL3b**, combined service signals from the HEU **130** through the fifth communication link **CL5b**, or amplified service signals from the EU **150** through the seventh communication link **CL7**.

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The RU processing system **145** may perform processes such as filtering, amplification, and the like for the received service signals in an analog way and/or digitally using radio resources allocated by the control of the RU controller **143**, and may transmit the processed service signals to an end-user device or another RU through a ninth communication link **CL9**.

The ninth communication link **CL9** may be a medium for transmitting analog or digital type service signals, for example, an RF cable, an optical fiber, an Ethernet-based cable, and the like. Although not shown in FIG. 2E, the RU processing system **145** may include a converter for converting the service signals received through the third communication link **CL3b**, the fifth communication link **CL5b**, and the seventh communication link **CL7** into signals suitable for processing therein and a converter for converting amplified signals to correspond to the ninth communication link **CL9**.

Referring to FIGS. 1 and 2F, the MSE **160** may include a management system entity interface **161** (hereinafter referred to as an MSE interface), a bus **163**, and a management system entity processing system **165** (hereinafter referred to as an MSE processing system).

The MSE interface **161** is for the MSE **160** to transmit and receive pieces of information necessary for spectrum sharing access to and from the SC **110**, the RSD **120**, the HEU **130**, and the RU **140**.

The MSE **160** may transmit and receive the pieces of information to and from the SC **110**, the RSD **120**, the HEU **130**, and the RU **140** connected to the MSE **160** through the first communication links **CL1a**, **CL1b**, **CL1c**, and **CL1d**, respectively, by using the MSE interface **161**.

As described above, the first communication links **CL1a**, **CL1b**, **CL1c**, and **CL1d** may be, for example, but are not limited to, the Internet, and may be any wired and/or wireless communication link such as WiMax, network optical fiber, an Ethernet-based cable, and the like.

The bus **163** may communicatively connect the MSE interface **161** to the MSE processing system **165**.

The MSE processing system **165** may include a processor **167** and a memory **169**.

The processor **167**, based on identification information of at least one RSD and a DAS received (or pre-saved) through the MSE interface **161** or information about radio access technology, may be any device suitable for executing program instructions for generating interworking information indicating an interworking status of the RSD and the DAS, virtualized RSD information representing the RSD and the DAS as a virtual unified RSD, or the like.

Alternatively, the processor **167** may be any device suitable for executing a program instruction for processing interworking information received from the RSD or the DAS through the MSE interface **161**, virtualized RSD information, or the like.

The processor **167** may be any device suitable for executing program instructions for monitoring, managing, controlling, and operating all operating states of RSDs **120**, or program instructions for monitoring, managing, controlling, and operating all operating states of the DAS.

The memory **169** may be any non-transitory medium for storing the program instructions described above that define an operation of MSE **160**. For example, the memory **169** may be ROM, RAM, an optical storage, a magnetic storage, a flash memory, or any other medium.

FIG. 3 is a block diagram of a spectrum sharing system according to an embodiment, and FIGS. 4 and 5 are flow-

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charts for illustrating a method of operating the spectrum sharing system shown in FIG. 3.

In more detail, the spectrum sharing system shown in FIG. 3 illustrates an embodiment in which the HEU 130 of a distributed antenna system interworks with the plurality of RSDs 120a and 120b, and the SC 110, the RSDs 120a and 120b, and the HEU 130 are communicatively connected to each other through the MSE 160 to transmit and receive information necessary for spectrum sharing access.

In FIG. 3, between the plurality of RSDs 120a and 120b and the HEU 130 and between the HEU 130 constituting a DAS and an RU and an EU, radio service signals provided to/from an end-user device are transmitted as analog or digital type signals. Processes related to configurations for this will not be given herein for convenience of explanation. This also applies to FIGS. 6 and 9 below.

In the description of FIGS. 3 to 5, the same or corresponding reference numerals as those in FIGS. 1 to 2F denote the same or corresponding elements, and therefore, repeated descriptions thereof will not be given herein. In the spectrum sharing system according to the present embodiment, allocation operations of shared radio resources for the RSDs 120a and 120b and the HEU 130 will be mainly described.

First, referring to FIGS. 1 to 2F, 3, and 4, in operation S401, the HEU 130 transmits operation information of the DAS (hereinafter referred to as DAS information) to the MSE 160.

The DAS information may include identifiers such as a DAS and/or an HEU constituting the DAS, an indication of radio access technology that the DAS may provide, operating parameters related to the radio access technology (e.g., synchronization, a slot structure, a silence interval, etc.), a geographical location (e.g., a concept related to an installation location of units or an installation location of an antenna connected to the units, and including a geographical area), and the like.

The operating parameters may include, for example, frequency spectrum information, level/power information, operating state information, and the like of a service signal conforming to the radio access technology.

According to an embodiment, the MSE 160 may be provided by a DAS manufacturer, and may be a network management system for the corresponding manufacturer to monitor and control an operating state of the DAS in real time. In this case, the MSE 160 may store DAS information in advance, and accordingly, operation S401 may be omitted, and the MSE 160 may perform the operations described later below based on the pre-stored DAS information.

According to an embodiment, the HEU 130 may transmit the DAS information to the MSE 160 offline.

In operation S402, the RSDs 120a and 120b transmit RSD information RSD #1 and RSD #2 to the MSE 160, respectively.

The RSD information RSD #1 and RSD #2 may include an indication of radio access technologies respectively provided by the RSDs 120a and 120b, operating parameters related to the radio access technologies, a geographic location, a device identifier, an available channel, an available frequency spectrum range, and the like.

According to an embodiment, the MSE 160 may be provided by a manufacturer of the RSDs 120a and 120b or a service provider that provides a spectrum sharing service through the RSDs 120a and 120b and the SC 110, and may be a network management system for the corresponding manufacturer or the service provider to monitor and control an operating state of the RSDs 120a and 120b in real time.

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In this case, the MSE 160 may store the RSD information in advance, and accordingly, operation S402 may be omitted, and the MSE 160 may perform the operations described later below based on the pre-stored RSD information.

According to an embodiment, the RSDs 120a and 120b may transmit the RSD information to the MSE 160 offline.

In operation S403, the MSE 160 generates interworking information of the DAS and the RSDs 120a and 120b based on obtained DAS information and the RSD information, and transmits the generated interworking information to the SC 110 in operation S404.

When the RSDs 120a, 120b operate as a signal source of the DAS, the DAS is subject to radio access technology of the RSDs 120a and 120b and operating parameters such as frequency bands related to the radio access technology.

In addition, due to the nature of an in-building wireless communication system, as a geographic location of the RSDs 120a and 120b, that is, their own location (or the location of antennas connected thereto), expands to the location of RUs of the DAS (or the location of antennas connected to the RUs), an indication of the geographic location of the RSDs 120a and 120b substantially coincides with an indication of the geographic location of the DAS.

Accordingly, the MSE 160, based on the DAS information and the RSD information, may generate the interworking information including an indication of an interworking state of the DAS and the RSDs 120a and 120b, an indication of radio access technologies available through the RSDs 120a and 120b and the DAS, operating parameters related to the radio access technologies, a geographic location, an available channel, information about an available frequency spectrum range, and the like. The interworking information may further include the DAS information and the RSD information.

In addition, the MSE 160 may transmit the interworking information to the SC 110 so that the SC 110 may recognize whether the RSDs 120a and 120b interwork with the DAS.

In operation S405, the SC 110 allocates shared radio resources to the RSDs 120a and 120b and the DAS, respectively, based on the received interworking information.

For example, after checking an interworking state of the RSDs 120a and 120b and the HEU 130 and recognizing the spectrum usage amount of priority users in a specific geographical location and/or a specific time set in which the RSDs 120a and 120b and the HEU 130 that interwork with each other are arranged, the SC 110 may allocate available shared radio resources to the RSDs 120a and 120b and the DAS, respectively, in consideration of respective geographical locations, operating states, frequency information, etc. of the RSDs 120a and 120b and the DAS.

According to an embodiment, the SC 110 may allocate radio resources such that shared radio resources allocated to the DAS include the shared radio resources respectively allocated to the RSDs 120a and 120b. This is because the DAS combines/distributes the radio resources of the RSDs 120a and 120b.

In operation S406, the SC 110 transmits allocation information indicating a result of the allocating to the MSE 160.

In operations S407 and S408, the MSE 160 transmits the received allocation information to the RSDs 120a and 120b and the HEU 130, respectively, and in operation S409, the RSDs 120a and 120b and the HEU 130 operate according to the received allocation information.

On the other hand, although not shown in FIG. 4, the HEU 130 transmits the allocation information received from the

MSE 160 to other elements of the DAS such as the RU 140 and the EU 150 so that the DAS may operate using the allocated radio resources.

According to certain aspects, the interworking information described above may be transmitted from the MSE 160 to the SC 110 as part of a registration process of the RSDs 120a and 120b and the HEU 130 for the SC 110. In addition, the interworking information may be transmitted from the MSE 160 to the SC 110 through at least one of periodic status updates of the RSDs 120a and 120b and the HEU 130 for the SC 110, or resource requests from the RSDs 120a and 120b and the HEU 130. In some cases, the periodic status updates may include radio environment measurements performed by at least one of the RSDs 120a and 120b and the HEU 130. This is also substantially the same in a method of allocating shared radio resources illustrated in FIGS. 6 to 8, and 9 to 11 later below.

Next, referring to FIGS. 1 to 2F, 3, and 5, in operation S501, the HEU 130 transmits the DAS information to the MSE 160.

As described above, when the MSE 160 is a network management system provided by the manufacturer of the DAS and the manufacturer monitors and controls a DAS operation state in real time, operation S501 may be omitted.

In operation S402, the RSDs 120a and 120b transmit the RSD information RSD #1 and RSD #2 to the MSE 160, respectively.

Likewise, when the MSE 160 is a network management system provided by the manufacturer of the RSDs 120a and 120b and the manufacturer monitors and controls the status of the RSDs 120a and 120b in real time, operation S502 may be omitted.

In operation S503, the MSE 160 generates virtualized RSD information (hereinafter referred to as VRSD information) based on the obtained DAS information and RSD information, and transmits the generated VRSD information to the SC 110 in operation S504.

The VRSD information may be information for allowing the SC 110 to recognize the DAS and the RSDs 120a and 120b as one integrated RSD.

The VRSD information may include information about an indication of radio access technology provided through the RSDs 120a and 120b and the DAS, operating parameters related to the radio access technology, a geographic location, a device identifier, an available channel, an available frequency spectrum range, and the like.

According to an embodiment, the VRSD information may include first VRSD information for recognizing the DAS as a device integrated with the RSD 120a or an extension device of the RSD 120a and second VRSD information for recognizing the DAS as a device integrated with the RSD 120b or an extension device of the RSD 120b.

In operation S505, the SC 110 allocates radio resources shared integrally to the RSDs 120a and 120b and the DAS, respectively, based on the received VRSD information.

For example, after recognizing the RSDs 120a and 120b and the HEU 130 as one virtualized RSD and recognizing the spectrum usage amount of priority users in a specific geographical location and/or a specific time set in which the RSDs 120a and 120b and the HEU 130 are arranged, the SC 110 may allocate available shared radio resources to the RSDs 120a and 120b and the HEU 130, integrally.

According to an embodiment, the SC 110 may recognize the RSD 120a and the HEU 130 as one virtualized RSD, may recognize RSD 120b and the HEU 130 as another

virtualized RSD, and may allocate shared radio resources to the RSD 120a and the HEU 130, and to the RSD 120b and the HEU 130, respectively.

In operation S506, the SC 110 transmits allocation information indicating a result of allocating the radio resources to the virtualized RSD to the MSE 160.

In operation S507, the MSE 160 determines respective operations of the RSDs 120a and 120b and the DAS based on the received allocation information. In operation S508, the MSE 160 transmits information about the determined operations for the RSDs 120a and 120b (hereinafter referred to as RSD operation information) to the RSDs 120a and 120b, respectively. In operation S509, the MSE 160 transmits information about the determined operation for the DAS (hereinafter referred to as DAS operation information) to the HEU 130.

In operation S510, each of the RSDs 120a and 120b operates according to the corresponding RSD operation information, and the HEU 130 operates according to the DAS operation information.

On the other hand, the HEU 130 transmits the DAS operation information to other elements of the DAS such as the RU 140 and the EU 150 so that the DAS may operate using the allocated radio resources.

FIG. 6 is a block diagram of a spectrum sharing system according to an embodiment, and FIGS. 7 and 8 are flowcharts for illustrating a method of operating the spectrum sharing system shown in FIG. 6.

In more detail, the spectrum sharing system shown in FIG. 6 illustrates an embodiment in which the HEU 130 of a DAS interworks with the RSDs 120a and 120b, and the SC 110 and the RSDs 120a and 120b are communicatively connected to each other through the MSE 160 to transmit and receive information necessary for spectrum sharing access.

In the description of FIGS. 6 to 8, the same or corresponding reference numerals as those in FIGS. 1 to 2F denote the same or corresponding elements, and therefore, repeated descriptions thereof will not be given herein. In the spectrum sharing system according to the present embodiment, allocation operations of shared radio resources for the RSDs 120a and 120b and the HEU 130 will be mainly described.

First, referring to FIGS. 1 to 2F, 6, and 7, in operation S701, the HEU 130 generates information of the DAS (hereinafter referred to as DAS information) and transmits the DAS information to any one of the RSDs 120a and 120b. In some embodiments, either RSD receiving the DAS information may function as a domain proxy.

The DAS information, which is information related to a radio service of the RSDs 120a and 120b provided to end-user devices through the DAS, may include, for example, radio access technologies respectively provided from the RSDs 120a and 120b to the node units (HEU, RU, and EU) of the DAS, operating parameters related to the radio access technologies, a geographic location, a device identifier, and the like.

According to an embodiment, when neither of the RSDs 120a and 120b operates as a domain proxy, the HEU 130 may transmit the DAS information to each of the RSDs 120a and 120b.

In operation S702, any one of the RSDs 120a and 120b generates interworking information of the RSDs 120a and 120b and the DAS based on the DAS information received from the HEU 130.

The interworking information, which is information directly or indirectly indicating whether or not the RSDs

120a and **120b** interwork with the DAS, may include an indication of an interworking state, an indication of radio access technology provided by the RSDs **120a** and **120b** through the DAS, operating parameters related to the radio access technologies, a geographic location, a device identifier, an available channel, an available frequency spectrum range, and the like.

In addition, because the DAS information includes all information such as the radio access technologies provided by the RSDs **120a** and **120b**, only one of the RSDs **120a** and **120b** may generate the interworking information based on the DAS information.

According to an embodiment, when the RSDs **120a** and **120b** do not function as a domain proxy, each of the RSDs **120a** and **120b** may receive the DAS information from the HEU **130** to generate interworking information.

In operation **S703**, any one of the RSDs **120a** and **120b** transmits the interworking information to the MSE **160**, and transmits its own information (hereinafter referred to as RSD information) and other RSD information to the MSE **160**. However, although not shown, the one RSD may also transmit the received DAS information to the MSE **160**.

According to an embodiment, when the RSDs **120a** and **120b** do not function as domain proxies, each of the RSDs **120a** and **120b** may transmit its own RSD information along with the generated interworking information to the MSE **160**.

In operation **S704**, the MSE **160** collects the received interworking information and the RSD information and transmits it to the SC **110**.

In operation **S705**, the SC **110** allocates shared radio resources to the RSDs **120a** and **120b** and the DAS, respectively, considering an interworking state based on the received interworking information and the RSD information.

In operation **S706**, the SC **110** transmits allocation information indicating a result of the allocating to the MSE **160**, in operation **S707**, the MSE **160** transmits the received allocation information to any one of the RSDs **120a** and **120b**, and in operation **S708**, the one RSD transmits the allocation information to the other RSD, and the RSDs **120a** and **120b** operate according to the received allocation information, respectively.

According to an embodiment, when neither of the RSDs **120a** and **120b** operates as a domain proxy, the MSE **160** may transmit the allocation information to each of the RSDs **120a** and **120b**.

In operation **S709**, the one of the RSDs **120a** and **120b** transmits the received allocation information to the HEU **130**, and in operation **S710**, the HEU **130** operates according to the received allocation information.

In addition, the HEU **130** transmits the allocation information received from the one RSD to other elements of the DAS, such as the RU **140** and the EU **150**, so that the DAS may operate using the allocated radio resources.

Next, referring to FIGS. **1** to **2F**, **6**, and **8**, in operation **S801**, the HEU **130** transmits DAS information to each of the RSDs **120a** and **120b**, and in operation **S802**, each of the RSDs **120a** and **120b** synthesizes its own information and the DAS information to generate individual virtualized RSD information (hereinafter referred to as individual VRSD information).

The individual VRSD information may be information that recognizes a DAS as a device integrated with any one of the RSDs **120a** and **120b** or an extension device for the any one, and may include an indication of radio access technology provided through any one of the RSDs **120a** and

120b and the DAS, operation parameters related to the radio access technology, a geographic location, a device identifier, and the like.

Subsequently, in operation **S803**, the RSDs **120a** and **120b** transmit the individual VRSD information to the MSE **160**, respectively, and in operation **S804**, the MSE **160** collects the individual VRSD information and transmits the individual VRSD information to the SC **110**.

Unlike the embodiments described with reference to FIG. **7** (embodiments in which any one of the RSDs may or may not operate as a domain proxy), the individual VRSD information is transmitted to the SC **110** through the MSE **160** so that the SC **110** recognizes the DAS connected to each of the RSDs **120a** and **120b** as an integrated (or extended) device.

In operation **S805**, the SC **110**, based on the received individual VRSD information, allocates radio resources shared integrally to the RSD **120a** and the DAS (i.e., one virtualized RSD), and the RSD **120b** and the DAS (i.e., another virtualized RSD), respectively.

In operation **S806**, the SC **110** transmits individual allocation information indicating a result of the respectively allocating radio resources to the virtualized RSDs to the MSE **160**, and in operation **S807**, the MSE **160** transmits the individual allocation information to the RSDs **120a** and **120b**, respectively.

In operation **S808**, the RSD **120a** determines respective operations of the RSD **120a** and the DAS based on the received individual assignment information, and the RSD **120b** determines respective operations of the RSD **120b** and the DAS based on the received individual allocation information.

In operation **S809**, each of the RSDs **120a** and **120b** operates according to a result of the determination.

In operation **S810**, each of the RSDs **120a** and **120b** transmits information about the determined operation of the DAS (hereinafter referred to as DAS operation information) to the HEU **130**. In operation **S811**, the HEU **130** operates according to the pieces of DAS operation information.

On the other hand, the HEU **130** transmits the pieces of DAS operation information to other elements of the DAS such as the RU **140** and the EU **150** so that the DAS may operate using the allocated radio resources.

FIG. **9** is a block diagram of a spectrum sharing system according to an embodiment, and FIGS. **10** and **11** are flowcharts for illustrating a method of operating the spectrum sharing system shown in FIG. **9**.

In more detail, the spectrum sharing system shown in FIG. **9** illustrates an embodiment in which the HEU **130** of a DAS interworks with the RSDs **120a** and **120b**, and the SC **110** and the HEU **130** are communicatively connected to each other through the MSE **160** to transmit and receive information necessary for spectrum sharing access.

In the description of FIGS. **9** to **11**, the same or corresponding reference numerals as those in FIGS. **1** to **2F** denote the same or corresponding elements, and therefore, repeated descriptions thereof will not be given herein. In the spectrum sharing system according to the present embodiment, allocation operations of shared radio resources for the RSDs **120a** and **120b** and the HEU **130** will be mainly described.

First, referring to FIGS. **1** to **2F**, **9**, and **10**, in operation **S1001**, the RSDs **120a** and **120b** respectively generate their own information (hereinafter referred to as RSD #1 information and RSD #2 information) and transmit the RSD #1 information and RSD #2 information to the HEU **130**.

The RSD #1 information and the RSD #2 information may include an indication of radio access technologies respectively provided by the RSDs 120a and 120b, operating parameters related to the radio access technologies, a geographic location, a device identifier, an available channel, an available frequency spectrum range, and the like.

In operation S1002, the HEU 130 generates interworking information of the plurality of RSDs 120a and 120b and the DAS based on the RSD #1 information and RSD #2 information respectively received from the plurality of RSDs 120a and 120b.

Subsequently, in operation S1003, the HEU 130 transmits the interworking information, the RSD #1 information and the RSD #2 information respectively received from the RSDs 120a and 120b, and its own information (hereinafter referred to as DAS information) to the MSE 160. As a domain proxy, the HEU 130 may transmit information about other interworking elements, for example, the RSDs 120a and 120b and the DAS, to the MSE 160 together with the interworking information.

In operation S1004, the MSE 160 collects the received interworking information and the like and transmits it to the SC 110.

In operation S1005, the SC 110 allocates shared radio resources to the RSDs 120a and 120b and the DAS, respectively, considering an interworking state based on the received interworking information and the like.

In operation S1006, the SC 110 transmits allocation information indicating a result of the allocating to the MSE 160, in operation S1007, the MSE 160 transmits the allocation information to the HEU 130, and in operation S1008, the HEU 130 transmits the received allocation information to the RSDs 120a and 120b.

In operation S1009, the HEU 130 and the RSDs 120a and 120b operate according to the allocation information.

Although not shown in FIG. 10, the HEU 130 transmits the allocation information received from the SC 110 to other elements of the DAS such as the RU 140 and the EU 150 so that the DAS may operate using the allocated radio resources.

Next, referring to FIGS. 1 to 2F, 9, and 11, in operation S1101, the RSDs 120a and 120b respectively transmit their own information (hereinafter referred to as RSD #1 information and RSD #2 information) to the HEU 130, and in operation S1102, the HEU 130 generates virtualized RSD information (hereinafter referred to as VRSD information) by synthesizing its own information and the RSD #1 information and RSD #2 information.

The VRSD information may be information that recognizes a DAS as a device integrated with the RSDs 120a and 120b or an extension device for the RSDs 120a and 120b, and may include radio access technology provided through the RSDs 120a and 120b and the DAS, operation parameters related to the radio access technology, a geographic location, a device identifier, an available channel, an available frequency spectrum range, and the like.

According to an embodiment, the VRSD information may include first VRSD information for recognizing the DAS as a device integrated with the RSD 120a or an extension device of the RSD 120a and second VRSD information for recognizing the DAS as a device integrated with the RSD 120b or an extension device of the RSD 120b.

In operation S1103, the HEU 130 transmits the VRSD information to the MSE 160, and in operation S1104, the MSE 160 transmits the VRSD information to the SC 110.

In operation S1105, the SC 110 allocates radio resources shared integrally to the RSDs 120a and 120b and the DAS, respectively, based on the received VRSD information.

In operation S1106, the SC 110 transmits allocation information indicating a result of allocating the shared radio resources to the MSE 160, and in operation S1107, the MSE 160 transmits the allocation information to the HEU 130.

In operation S1108, the HEU 130 determines respective operations of the RSDs 120a and 120b and the DAS based on the received allocation information.

In operation S1109, the HEU 130 operates according to a result of the determination of the DAS.

The HEU head-end unit 130 transmits information about the result of the determination to other elements of the DAS such as the RU 140 and EU 150 so that the DAS may operate using the allocated radio resources.

In operation S1110, the HEU 130 transmits information about the determined operations of the RSDs 120a and 120b (RSD #1 operation information and RSD #2 operation information) to the RSDs 120a and 120b. Accordingly, in operation S1111, the RSDs 120a and 120b operate according to the received operation information, respectively.

FIGS. 3 to 11 describe the embodiment in which the HEU 130 interworks with the plurality of RSDs 120a and 120b above. However, even in an embodiment in which the RU 140 interworks with at least one RSD, the allocation operation of the shared radio resources as shown in FIGS. 3 to 11 will be possible.

Further, FIGS. 4, 5, 7, 8, 10, and 11 and the methods described with reference thereto include one or more operations and/or actions for achieving the methods. The operations and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, the order and/or use of specific operations and/or actions may be modified without departing from the scope of the claims, unless a certain order for the operations and/or actions is specified.

In addition, various operations of the methods described above may be performed by any suitable means capable of performing corresponding functions. The means includes, but is not limited to, various hardware and/or software components and/or modules such as an application specific integrated circuit (ASIC) or a processor. In general, when there are operations corresponding to the drawings, these operations may have a corresponding counterpart and functional components having the same number as the number of the counterpart.

The various illustrative logic blocks, modules, circuits, and processors described in connection with the disclosure may be implemented or performed by a general-purpose processor designed to perform the functions disclosed herein, a digital signal processor (DSP), an ASIC, a field-programmable gate array (FPGA) or other programmable logic device (PLD), a discrete gate or transistor logic device, discrete hardware components, or any combination thereof. The general-purpose processor may be a microprocessor, but may alternatively be any commercially available processor, controller, microcontroller, or state machine. The processor may also be implemented in a combination of computing devices, for example, a combination of the DSP and the microprocessor, a plurality of microprocessors, one or more microprocessors in connection with a DSP core, or any other configuration.

According to embodiments of the disclosure, when a spectrum sharing system interworks with a distributed antenna system, a system controller of the spectrum sharing system communicates with radio service devices of the

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spectrum sharing system and node units of the distributed antenna system through a management system entity.

Accordingly, as the spectrum sharing system is able to minimize a management and control burden of the system controller and efficiently allocate and operate shared radio resources by considering whether to interwork with the distributed antenna system, and the distributed antenna system interworks with the spectrum sharing system, it is possible to effectively prevent unexpected interference from occurring in a specific area and/or at a specific time.

Effects obtainable by the disclosure are not limited to the effects described above, but other effects not described herein may be clearly understood by one of ordinary skill in the art from the above descriptions.

Numerous modifications and adaptations will be readily apparent to one of ordinary skill in the art without departing from the spirit and scope of the disclosure.

In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein.

While the disclosure has been particularly shown and described with reference to embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

What is claimed is:

1. A method of interoperating a distributed antenna system (DAS) with a spectrum sharing system (SSS), the method comprising:

generating, by a management system entity (MSE), inter-operating information based on DAS information and radio service device (RSD) information;

transmitting, by the MSE, the interoperating information to a system controller of the SSS, wherein the system controller of the SSS is configured to control spectrum sharing in the SSS; and

receiving, by the MSE, allocation information including a result of allocating shared radio resources to the DAS and at least one RSD, respectively, according to the interoperating information from the system controller.

2. The method of claim 1, wherein the interoperating information comprises information about at least two of an indication of an interoperating state of the at least one RSD and the DAS, an indication of radio access technology (RAT) provided by the at least one RSD through the DAS, operation parameters related to the RAT, a geographic location, an available channel, and an available frequency spectrum range.

3. The method of claim 1, further comprising:
before the generating of the interoperating information, receiving, by the MSE, the DAS information from a node unit of the DAS; and

receiving, by the MSE, the RSD information from the at least one RSD.

4. The method of claim 1, wherein the transmitting of the interoperating information comprises transmitting, by the MSE, the interoperating information to the system controller as part of a registration process for the system controller of the DAS and the at least one RSD.

5. The method of claim 1, wherein the transmitting of the interoperating information comprises transmitting, by the MSE, the interoperating information to the system controller through at least one of a resource request to the system controller or periodic status update of the DAS and the at least one RSD.

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6. The method of claim 1, further comprising:
after the receiving of the allocation information, transmitting, by the MSE, the allocation information to a node unit of the DAS; and

transmitting, by the MSE, the allocation information to the at least one RSD.

7. The method of claim 1, wherein the MSE is configured to operate and manage the DAS by controlling at least one node unit of the DAS.

8. The method of claim 7, wherein the at least one node unit of the DAS is a head-end unit of the DAS communicatively connected to the at least one RSD, or a remote unit of the DAS communicatively connected to the at least one RSD.

9. The method of claim 1, wherein the MSE is configured to control an operation of the at least one RSD to operate and manage the SSS.

10. A method of interoperating a distributed antenna system (DAS) with a spectrum sharing system (SSS), the method comprising:

generating, by a management system entity (MSE), virtualized radio service device (RSD) information based on DAS information and RSD information;

transmitting, by the MSE, the virtualized RSD information to a system controller of the SSS, wherein the system controller of the SSS is configured to control spectrum sharing in the SSS;

receiving, by the MSE, allocation information including a result of allocating radio resources integrally shared to the DAS and the at least one RSD according to the virtualized RSD information from the system controller; and

determining, by the MSE, an operation of the DAS based on the allocation information.

11. The method of claim 10, wherein the virtualized RSD information is information that causes the system controller of the SSS to recognize the at least one RSD and the DAS as one device or the DAS as an extension device of the at least one RSD.

12. The method of claim 10, wherein the virtualized RSD information comprises information about at least two of an indication of a radio access technology (RAT) integrally supported by the at least one RSD and the DAS, operation parameters related to the RAT, a geographic location, an available channel, and an available frequency spectrum range.

13. The method of claim 10, wherein the determining of the operation of the DAS comprises determining, by the MSE, an operation of the at least one RSD based on the allocation information.

14. The method of claim 10, wherein the transmitting of the virtualized RSD information comprises transmitting, by the MSE, the virtualized RSD information to the system controller as part of a registration process for the system controller of the DAS and the at least one RSD.

15. The method of claim 10, wherein the transmitting of the virtualized RSD information comprises transmitting, by the MSE, the virtualized RSD information to the system controller through at least one of a resource request to the system controller or periodic status update of the DAS and the at least one RSD.

16. A method of interoperating a distributed antenna system (DAS) with a spectrum sharing system (SSS), the method comprising:

receiving, by a management system entity (MSE), one of interoperating information and virtualized radio service device (RSD) information from a node unit of the DAS or at least one RSD;

transmitting, by the MSE, the interoperating information or the virtualized RSD information to a system controller of the SSS;

receiving, by the MSE, allocation information including a result of allocating shared radio resources according to the interoperating information or the virtualized RSD information from the system controller; and

transmitting, by the MSE, the allocation information to the node unit of DAS and the at least one RSD.

17. The method of claim **16**, wherein the interoperating information or the virtualized RSD information is generated by one of the node unit of the DAS and the at least one RSD.

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